Part	Amount	Description / Type	Manufacturer	Supplier	Price / pie	Budgeted
Reservoir						
Columns	12	Spectra/Chrom® Organic Chromatography Columns, Replacement Column and Collars only, 5.0 cm ID X 60 cm long	Spectrum Chromatography	graphy, Houston, TX 77060, USA	1350.00	16200.00
Sealing O rings	24	O-Ring FPM 75, 78 * 6 mm, mid hard, 27482	arcus Dichtelemente	arcus Dichtelemente	95.00	2280.00
PEM Cell						
Power supply	12	Voltcraft PPS-11810	Voltcraft	Conrad	214.00	2568.00
Anodes	17	2.5-mm thick, 50 x 50 mm sintered DSE anode for O2 Evolution	DeNora Permelec	DeNora Permelec	540.00	9180.00
Cathodes	17	2.5-mm thick, 50 x 50 mm sintered DSE cathode for O2/H2 Evolution	DeNora Permelec	IAEA / Eawag workshop	520.00	8840.00
Export Fee for Denora	1		DeNora Permelec	DeNora Permelec	500.00	500.00
Nafion membrane	4	Nafion 117 sheet, pieces manually cut to 5 x 5 cm per cell (0.18 mm thickness, >0.90 meq/g exchange capacity	Dupont	https://www.alfa.com	114.00	456.00
Gold mesh	16	5 x 5 cm piece per cell, 99 % plain weave, gold mesh (0.88 grams), 0.25 mm aperture, 0.06 mm wire diameter, 65 % open screen area	Variable Manufacturers	Goodfellow	1894.00	30304.00
Gold wire	1	0.75mm thick	Variable Manufacturers	Goodfellow	2080.00	2080.00
Stainless steel mesh	4	5 x 5 cm piece per cell, SS316, Mesh #84, 0.239 mm aperture, 0.063 mm wire diameter, 65 % open screen area	Variable Manufacturers	The Mesh Company, UK	23.00	92.00
PTFE Cell (Base)	12	PTFE, CNC milled	custom made	Thomas Schiefer	264.76	3177.17
PTFE Cell (Top)	12	PTFE, CNC milled	custom made	Thomas Schiefer	248.72	2984.61
PTFE channel block	12	PTFE, CNC milled	custom made	Thomas Schiefer	72.21	866.50
PTFE pressure block	12	PTFE, CNC milled	custom made	Thomas Schiefer	54.56	654.69
Aluminium press ring	12				72.00	864.00
Sample outlet valve	12	Teflon ball valve, TEFLON VALVE, .080" THRU, 1/4" MFIT CONN	Repligen	Repligen	215.00	2580.00
Sealant “glue”	3	Loctite SI 5970			23.00	69.00
Condensor Top						
Power supply	12	Adjustable PSU	Voltcraft	Conrad.ch	245.95	2951.40
Alum Top	12	aluminium anodized	Thomas Schiefer	Thomas Schiefer	21.40	256.74
Peltier elements	24	Cooling element to facilitate condensation	Variable Manufacturers	Thomas Schiefer	21.40	513.48
plug and cables for cell	24	Connectors	Variable Manufacturers	Thomas Schiefer	10.70	256.74
ABS plastic top assembly for Column (two halves)	12	3D printed	Thomas Schiefer	Thomas Schiefer	41.08	492.94
ABS Top Mount						
Screws + nuts	80	Connectors	Variable Manufacturers	Thomas Schiefer	1.07	85.58
Insulation	12	Important for efficiency	Variable Manufacturers	Thomas Schiefer	4.28	51.35
1/8" Swagelok connector	12	Connectors	Swagelok	Thomas Schiefer	12.84	154.04
Rack						
Column hanger (upper)	4	milled out of a 10mm board	custom made	Eawag workshop	2.00	8.00
tube supplier (lower)	4	milled out of a 10mm board	custom made	Eawag workshop	3.00	12.00
aluminium frame	1	Cut and connectors using standard framing	custom made	Eawag workshop	120.00	120.00
Extraction Loop						
Heating element	2	Standard Laboratory Apparatus	Variable Manufacturers	Fischer Scientific CH	100.00	200.00
Erlenmeyer Flask	2	Standard Laboratory Apparatus	Variable Manufacturers	PharmaGlas	36.00	72.00
Stainless Tubing	8	Standard Laboratory Apparatus	Variable Manufacturers	HuberLab	32.00	256.00
Capture Flask (Cold Trap)	2	Standard Laboratory Apparatus	Variable Manufacturers	HuberLab	68.00	136.00
Pump	12	To evacuate the cells completely after enrichment	KNF Pumps	KNF Pumps	3123.70	37484.40
Glass Tubing	8	To prevent water loss and/or fractionation	Variable Manufacturers	PharmaGlas	40.00	320.00
Note: Construction of all devices is typical for a laboratory run by physicists, where method and analytical tools are continuously developed. W+T's in-kind contribution to build the cells in-house will lead to considerable cost reduction and the build-up of expertise on the construction and operation of such advanced electrochemical enrichment cells.					Total	**121420.21**
					VAT (8.1%)	**9835.04**
					Total+VAT	**131255.25**